

UMAN', Ukraine

WMO: 335870

Lat: 48.7669N

Lon: 30.2328E

Elev: 216

StdP: 98.76

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-14.6	-20.9	0.6	-17.7	-17.8	0.8	-13.9	12.7	-3.3	12.1	-3.5	1.1	320	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.6	20.4	29.9	19.9	28.3	19.1	21.7	28.8	20.9	27.6	20.1	26.3	1.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.7	24.7	18.7	13.9	23.9	17.9	13.2	23.1	64.5	29.0	61.5	27.6	58.7	26.3	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.8	8.2	5.4	DB	-20.8	33.9	3.6	2.0	-23.4	35.3	-25.5	36.5	-27.6	37.6	-30.2	39.1
(5)				WB	-21.1	23.3	3.6	1.7	-23.6	24.6	-25.7	25.6	-27.7	26.5	-30.3	27.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-3.9	-2.3	2.5	9.7	15.3	18.8	20.9	20.1	14.6	8.3	3.2	-1.9	(6)
(7)		DBStd	9.80	5.57	5.54	4.52	4.00	3.98	3.32	2.88	3.30	3.84	4.29	4.79	5.38	(7)
(8)		HDD10.0	1737	432	345	235	52	6	0	0	0	7	83	209	370	(8)
(9)		HDD18.3	3704	690	578	491	259	109	34	9	18	122	310	454	628	(9)
(10)		CDD10.0	1312	0	0	2	43	170	264	339	313	144	31	4	0	(10)
(11)		CDD18.3	239	0	0	0	1	16	49	89	73	10	1	0	0	(11)
(12)		CDH23.3	2441	0	0	0	13	201	471	834	786	132	4	0	0	(12)
(13)		CDH26.7	741	0	0	0	2	41	119	268	283	29	0	0	0	(13)
(14)	Wind	WSAvg	1.8	2.0	2.1	2.4	2.0	1.7	1.6	1.4	1.5	1.5	1.6	1.8	1.8	(14)
(15)	Precipitation	PrecAvg	630	45	38	37	46	53	88	84	58	51	38	46	44	(15)
(16)		PrecMax	1000	177	103	127	112	150	170	210	168	193	125	95	94	(16)
(17)		PrecMin	428	7	3	2	10	3	11	15	6	1	3	11	7	(17)
(18)		PrecStd	122	32	25	26	23	35	41	48	34	47	31	24	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	10.3	17.3	25.1	30.3	31.5	33.5	34.5	29.6	23.3	16.6	10.2	(19)
(20)			MCWB	5.1	6.7	9.7	14.9	18.5	20.2	21.7	21.2	18.7	15.7	12.4	8.5	(20)
(21)		2%	DB	5.4	7.4	13.9	21.8	27.6	29.6	31.4	32.1	26.9	20.0	13.4	8.2	(21)
(22)			MCWB	3.9	5.0	8.0	12.9	17.3	19.7	20.7	20.4	17.5	14.4	11.0	7.0	(22)
(23)		5%	DB	3.8	5.6	11.3	19.6	25.6	28.0	29.7	29.7	24.4	17.1	11.3	6.4	(23)
(24)			MCWB	2.7	3.9	6.8	11.8	16.7	19.1	20.3	19.7	16.2	13.1	9.7	5.5	(24)
(25)		10%	DB	2.5	4.0	9.1	17.4	23.5	26.2	27.8	27.7	22.0	14.8	9.8	4.6	(25)
(26)			MCWB	1.7	2.7	5.6	10.7	15.6	18.6	19.6	18.8	15.4	11.8	8.5	3.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.5	10.2	15.9	19.7	22.3	23.1	22.8	20.2	17.1	13.1	9.1	(27)
(28)			MCDWB	6.6	9.2	16.0	24.7	26.6	28.0	29.9	31.3	26.4	21.9	15.1	9.7	(28)
(29)		2%	WB	4.1	5.4	8.6	13.7	18.4	21.0	21.9	21.5	18.4	15.1	11.4	7.1	(29)
(30)			MCDWB	5.2	7.1	12.8	20.2	25.4	27.2	28.7	29.3	24.6	18.5	13.2	7.9	(30)
(31)		5%	WB	2.9	4.1	7.3	12.6	17.5	20.1	21.2	20.5	17.1	13.6	10.0	5.5	(31)
(32)			MCDWB	3.6	5.3	10.7	18.1	23.8	26.1	27.7	27.3	22.9	16.8	11.2	6.3	(32)
(33)		10%	WB	1.7	2.7	5.9	11.4	16.5	19.2	20.3	19.6	15.9	12.1	8.6	3.9	(33)
(34)			MCDWB	2.4	3.9	8.7	16.3	22.2	24.7	26.4	25.9	20.6	14.5	9.7	4.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	5.5	7.5	10.3	11.5	10.8	11.0	11.8	10.4	8.7	5.2	4.7	(35)
(36)			MCDBR	4.7	7.0	11.8	14.3	14.5	13.6	13.8	15.2	15.0	12.9	7.5	5.6	(36)
(37)			MCWBR	4.0	5.1	7.1	7.0	6.1	5.5	5.2	5.5	7.0	7.5	5.1	4.7	(37)
(38)		5% WB	MCDBR	4.2	6.2	10.6	12.3	12.5	11.7	12.2	12.9	13.0	11.2	6.6	5.2	(38)
(39)			MCWBR	3.7	5.0	6.9	6.5	5.7	5.4	5.2	5.2	6.9	7.0	4.9	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.344	0.378	0.435	0.407	0.402	0.430	0.432	0.395	0.365	0.344	0.314	(40)	
(41)		taud	2.235	2.192	2.218	2.119	2.273	2.306	2.233	2.225	2.296	2.364	2.334	2.298	(41)	
(42)		Ebn at Noon	724	786	824	810	854	860	827	806	800	762	688	674	(42)	
(43)		Edh at Noon	77	102	117	144	128	125	133	128	108	86	71	65	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.99	1.80	3.03	4.46	5.77	6.16	5.99	5.35	3.64	2.21	0.94	0.70	(44)	
(45)		RadStd	0.18	0.32	0.43	0.43	0.47	0.42	0.35	0.31	0.54	0.31	0.13	0.17	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.59	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	+116	N/A
(47)	Regional (9 neighbors)	+0.69	N/A	N/A	+0.95	N/A	N/A	-129	-185	+111	+54

Nomenclature: See separate page